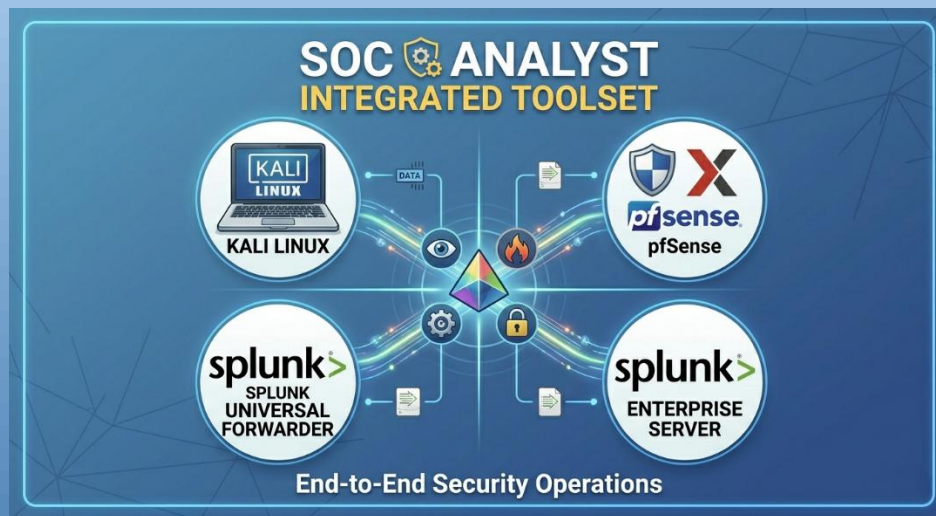




PROJECT:

Design and Implementation of a Security Operations Center (SOC) to Monitoring System Using pfSense Firewall and Splunk Enterprise Tool used in project



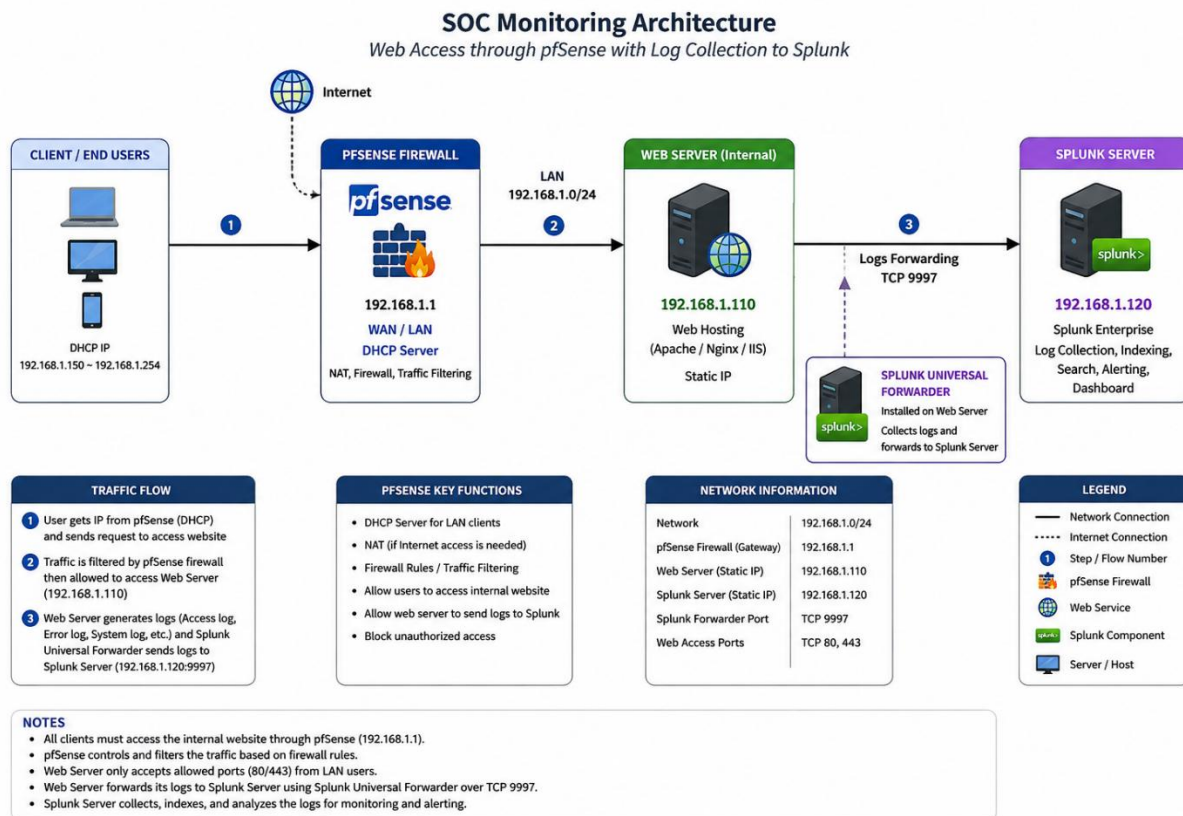
SOC Monitoring Project Implementation Report

1. Project Information

Project Title

SOC Monitoring and Web Access Log Collection Using pfSense and Splunk

PROJECT Architecture:



Project Duration: 4 Weeks

Project Team

Name	Role
CHHON SOCHEAT	Project Lead & Solution Architect
PENG SEYHA	Network & Firewall Engineer

PENCHANTHAN VITHATTHAI	System & Web Server Engineer
PUM CHAMNAN	SOC Analyst & SIEM Engineer

2. Executive Summary

This project aims to design and implement a Security Operations Center (SOC) monitoring solution for a web application environment. The solution utilizes pfSense Firewall as the network security gateway and Splunk Enterprise as the Security Information and Event Management (SIEM) platform.

The architecture enables users to access an internal web application through pfSense while collecting and forwarding security and access logs from the web server to Splunk for monitoring, analysis, alerting, and incident investigation.

3. Project Objectives

The objectives of this project are:

1. Deploy a secure network environment using pfSense Firewall.
 2. Host an internal web application accessible through controlled network access.
 3. Collect web access and system logs using Splunk Universal Forwarder.
 4. Centralize log management using Splunk Enterprise.
 5. Develop SOC monitoring dashboards and security alerts.
 6. Demonstrate security event monitoring and incident investigation capabilities.
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4. Project Scope

In Scope

- pfSense Firewall deployment
- DHCP and Network Configuration
- Firewall Rule Configuration
- Internal Website Deployment

- Splunk Enterprise Installation
- Splunk Universal Forwarder Deployment
- Log Collection and Analysis
- Dashboard and Alert Development
- Security Monitoring Demonstration

Out of Scope

- External Internet Exposure
- Advanced Threat Detection
- High Availability (HA) Configuration
- Disaster Recovery Site

5. System Architecture

Network Components:

Component	IP Address	Purpose
pfSense Firewall	192.168.1.1	Gateway, DHCP, Security Filtering
Web Server	192.168.1.110	Internal Website Hosting
Splunk Server	192.168.1.120	Log Collection and SIEM
End Users	DHCP	Access Internal Website

Traffic Flow:

1. Users access the website through pfSense Firewall.
2. pfSense inspects and filters incoming traffic.
3. Requests are forwarded to the Web Server.
4. Web Server generates access and system logs.
5. Splunk Universal Forwarder collects logs.
6. Logs are forwarded to Splunk Enterprise via TCP Port 9997.

7. Splunk indexes data and generates dashboards and alerts.
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6. Team Responsibilities

CHHON SOCHEAT – Project Lead

Responsibilities:

- Project planning and coordination
- Architecture design
- Progress tracking
- Integration testing
- Documentation review
- Final presentation and reporting

PENCHANTHAN VITHATTHAI – Network & Firewall Engineer

Responsibilities:

- Install and configure pfSense
- Configure DHCP services
- Implement firewall policies
- Configure NAT and traffic filtering
- Validate network security controls

PENG SEYHA – System & Web Server Engineer

Responsibilities:

- Deploy server infrastructure
- Configure web services
- Deploy sample website
- Install Splunk Universal Forwarder
- Generate application logs

PUM CHAMNAN – SOC Analyst

Responsibilities:

- Install Splunk Enterprise
 - Configure data ingestion
 - Create dashboards and alerts
 - Analyze security events
 - Demonstrate incident investigation workflow
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7. Project Schedule

Week 1 – Infrastructure Deployment

- Project kickoff
- Architecture design
- pfSense installation
- Web server deployment
- Splunk server installation

Week 2 – Configuration Phase

- Configure DHCP and firewall policies
- Deploy website
- Configure Splunk receiving port
- Establish log forwarding connectivity

Week 3 – Monitoring Implementation

- Validate log collection
- Create dashboards
- Configure alerts
- Conduct integration testing

Week 4 – Validation and Reporting

- End-to-end testing

- Incident simulation
 - Final documentation
 - Presentation preparation
 - Project demonstration
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8. Deliverables

Network Deliverables

- pfSense Firewall Configuration
- DHCP Configuration
- Firewall Rule Documentation

Server Deliverables

- Web Server Deployment
- Website Accessibility Verification

SIEM Deliverables

- Splunk Enterprise Deployment
- Log Collection Configuration
- Monitoring Dashboard
- Security Alert Rules

Documentation Deliverables

- Architecture Diagram
 - IP Address Plan
 - Configuration Guide
 - Test Results
 - Final Project Report
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9. Risk Assessment

Risk	Impact	Mitigation
Firewall Misconfiguration	High	Peer Review and Testing
Log Collection Failure	High	Verify Forwarder Connectivity
Server Downtime	Medium	Backup Configuration
Schedule Delay	Medium	Weekly Progress Review

10. Expected Outcome

Upon completion, the project will provide a functional SOC monitoring environment capable of collecting, analyzing, and monitoring web access logs through Splunk. The solution will demonstrate practical cybersecurity operations including log management, alerting, dashboard visualization, and incident investigation.